MAKEGOOD POWER CHAIR

Decision Matrix and Regulatory Notes

Options per subsystem, trade-offs, and compliance flags

R.O.A.R. / TSSFAA with **MakeGood INC**
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Private pre-release draft. Not yet safe to build from. Prices are September 2026 US retail.

Pre-purchase decisions a builder makes before ordering parts. Each row is a choice; the configurator (`/configurator`) encodes the same choices and resolves them into a parts list with pricing. This doc is the human-readable record of *why* each option exists and what it trades off.

Status: **DRAFT.** Options listed are candidates, not endorsements. Cost figures are order-of-magnitude placeholders until the BOM is validated.

Legend: ✅ recommended default · ⚠️ caution / needs verification · ⛔ blocking combination

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1. Drive motors

Option	Pros	Cons	Notes
Donated hoverboard hubs (teardown)	$0, abundant, built-in tire	Variable quality; axle/hall variants; must pass incoming inspection	Controller always replaced. See inspection checklist below.
New hoverboard-class hubs	Known-good, consistent	~$40–80/pair	Specify one axle variant in BOM
Standard 24 V wheelchair motors (gearmotor + wheel)	Integrated brake, wheelchair-rated, proven	Heavier, pricier, larger	Pairs naturally with standard wheelchair controllers

Teardown incoming inspection: bearings free/no play · all 6 hall states valid · matched phase resistance L/R · insulation check · tire condition · axle flats undamaged. Recalibrate controller after any motor swap.

2. Motor controller

Requirements: configuration-only (no reflashing) · closed-loop speed control (no downhill runaway) · current limits · command-loss watchdog · brake resistor / regen handling · fault reporting · versioned config file.

Option	Fit	Notes
VESC dual	✅ best open-source fit	FOC with halls; config exports from VESC Tool. Clone quality varies — specify vetted units.
ODrive S1 ×2	✅	12–50.5 V, 40 A cont. w/ heat spreader; USB/UART/CAN/PWM/analog; ships brake resistor; publishes hoverboard setup guide. ⚠️ At 42 V full charge only ~8 V headroom to trip — brake resistor mandatory.
RoboClaw	✅	Simple, robust, well documented
Sabertooth	⚠️	Brushed-oriented; limited for hub motors

Option	Fit	Notes
Reflashed hoverboard board (EFeru FOC)	⚠️	Cheap but violates no-reflash rule; not reproducible for volunteers
Standard wheelchair controller	✅ with wheelchair motors	⛔ with hoverboard hubs (brushless mismatch)

Supervisor ↔ controller interface: CAN preferred, UART acceptable; checksum + timeout required.

3. Battery

Option	Pros	Cons
DeWalt 20V MAX packs (hot-swap pair)	Ubiquitous, cheap, parents already own them, airline-friendly sizes	Needs adapter + ideal-diode ORing for hot swap; 18 V nominal limits top speed
Airline-allowed power banks	Travel-friendly	Current limits usually too low for drive — ⚠️ verify
24 V chair pack	Right voltage, wheelchair-rated	Heavier, not quick-swap by default
Custom pack + BMS	Fits exactly	Build complexity; ⚠️ FDA/UN 38.3 implications

⛔ Shared battery with a ventilator: see `regulatory-notes.md` before choosing.

4. Parking brake

Option	Notes
Manual wheel lock (TMT style)	Simplest; child/parent operated
Solenoid pin lock	Electrically released; fails-locked
Spring-applied EM brake	Wheelchair standard; fails-locked; power to release
Integrated wheelchair-motor brake	Comes with option 1c motors

Safety Floor rule: loss of power SHALL NOT release a brake on a slope.

5. Manual ↔ power mechanism

Option	Notes
Lever-deployed hub motors lifting printed wheels	Both modes on one frame; mechanical complexity

Option	Notes
Removable power pod	Clean separation; needs quick-connect
Friction drive on printed wheels	Cheapest; wear and wet-performance concerns

6. Passenger / buddy riding

Trailer · buddy seat · standing board · shared-play modes. ⛔ No passengers when a ventilator is mounted.

7. Supervision mode (parent-selected)

Dead-man switch · wired/wireless e-stop · tether · timed sessions · speed cap · attendant remote with kill switch · geofence (with assist). Each is a *configuration*, documented with trade-offs. Only the Safety Floor (device-fault protections) is fixed.

8. AI assist (optional, advisory)

Collision · drop-off/stair · street/vehicle · tip · power monitoring. Subtract-only: warn → slow → stop-with-override. Override method matches the driver's own input device. Candidate platforms: Arduino UNO Q / VENTUNO Q, or sponsor hardware.

Open decisions

- [ ] Lock BOM to one hoverboard axle variant, or design mount for all known variants
- [ ] Bench-test VESC dual vs 2× ODrive S1 on hoverboard hubs at chair loads
- [ ] Catalog hoverboard motor variants (wheel size, axle, pole pairs, hall connector)
- [ ] Settle regulatory position (trainer / DIY vs. FDA Class II powered wheelchair, 21 CFR 890.3860)
- [ ] License selection for public release

Regulatory Notes

Working notes, not legal advice. Flags to resolve before public release.

Classification

- **Powered wheelchair** — FDA Class II, 21 CFR 890.3860. Applies if the device is marketed as a wheelchair for a person with a disability.
- **Mobility trainer / DIY / toy** — the TMT itself sits outside medical-device regulation by being a published open design, not a marketed product. The powered add-on may or may not inherit that position. **[TBD — decide and document.]**

Standards to review for applicability

Standard	Topic
ISO 7176-1 / -2	Static / dynamic stability
ISO 7176-8	Strength
ISO 7176-14	Power and control systems
ISO 7176-21	EMC
IEC 62133-2	Li-ion cell/pack safety
UN 38.3	Li-ion transport
IATA DGR	Mobility-aid battery provisions (air travel)

Ventilator accommodation

- Mounting a ventilator on the chair is in scope. **Sharing the chair's drive battery with a ventilator is a separate question** — it places the battery in the ventilator's power path and likely pulls it into the ventilator's regulatory envelope.
- Default position until resolved: ventilator uses its own manufacturer-supplied battery. The chair may provide a *mount* and strain relief, not power.
- ⛔ No passengers / buddy riding with a ventilator mounted.

Liability posture for AI assist

- Advisory and subtract-only by design. Never commands motion or steering.
- Independent of the Safety Floor — a failed assist module must not degrade basic chair safety.
- Document clearly that assist is a convenience, not a safety certification.